SAI SREENIVAS CHINTHA

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Education

University Of Massachusetts Amherst | *M.S. in Computer Science* | GPA: **4.0/4.0** — **Sept 2024 - May 2026**
***Coursework**: Adv. NLP, Information Retrieval, Adv. Machine Learning, Responsible AI*

Indian Institute of Technology Hyderabad | *Bachelor's in Computer Science* — **Aug 2017 - Jun 2021**
***Coursework**: Machine Learning, Data Mining, Computer Architecture, Operating Systems*

Publications

- ***Reducing PII Leakage in Chain-of-Thought Reasoning***, ICML 2026 Agents in the Wild Workshop.
- ***XLM: A Python package for non-autoregressive language models***, EACL 2026 Demo.

Research Experience

Microsoft — **Amherst, MA**
Research Extern — ***Feb 2026 - Present***

- Developing an **entropy-guided adaptive decoding** method for **speculative decoding** approaches like **EAGLE** enabling dynamic token generation lengths and eliminating verification overhead while preserving output quality.
- Conducted ∼**1B token pretraining on PILE** and **instruction tuning on ShareGPT** datasets using **LLaMA-3 8B**.

UMass IESL Lab — **Amherst, MA**
Graduate Student Researcher, Advisor: Prof. Andrew McCallum — ***Oct 2025 - Present***

- Contributed to the development of the **XLM framework**, designed to ease the development of non-autoregressive LMs.
- Exploring a self-correction framework for diffusion language models to address uncoordinated parallel sampling of tokens.

KLA — **Milpitas, CA**
Machine Learning Research Intern — ***Jun 2025 - Aug 2025***

- Researched **zero-shot defect retrieval & detection** using vision–language models on SEM wafer images.
- Enhanced **Grounding DINO's** architecture with a **softmax-based** loss and **fine-tuned the text encoder (BERT)**.
- Implemented data augmentations including **Mosaic, targeted region cropping**, **diverse text prompt formulations**.

UMass CIIR Lab — **Amherst, MA**
Graduate Student Researcher — ***Feb 2025 - May 2025***

- Explored scalable personalized multimodal retrieval by integrating **VLMs** with multimodal language model-based re-ranking, leveraging **prompt engineering** and **Unsloth** for efficient fine-tuning on large instruction-tuned models.

Projects

Mitigating PII Leakage in Chain-of-Thought of LRMs | *UMass* — **Sept 2025 - Dec 2025**

- Analysed and reduced personally identifiable information (PII) leakage in large reasoning models.
- Performed **prompt engineering** and **LoRA-SFT** on **GPT-OSS-20B, LLaMA-3.3-70B** and **DeepSeek-R1-Qwen**.

Learned Sparse Retrieval using Vector Quantization | *UMass* — **Feb 2025 - April 2025**

- Engineered a novel learned sparse retrieval model using **vector quantization** with top-k sparsification on 2M samples of the **MS MARCO**. Optimized **Contriever** with contrastive, vector codebook, sparsity objectives.
- Outperformed **BM25** and sparse Contriever baselines on ∼7.5k passages, achieving **MRR@10** of **0.89**.

Fit-Aware Fashion Recommendation Re-ranking | *UMass* — **Sept 2024 - Nov 2024**

- Designed a re-ranking system by fine-tuning **CLIP** and extracting features using **Faster R-CNN** and **K-means**.
- Achieved Recall@1 of **0.42** & Recall@20 of **0.95**, significantly outperforming baseline models.

Layout Generation Using Diffusion model | *Adobe* — **Dec 2023 - Jan 2024**

- Developed a diffusion model for generating aesthetic layouts with text, images, and shapes.
- Applied continuous and discrete diffusion with conditional generation, achieving an **FID** score of **15.2**.

Work Experience

Adobe — **Bengaluru, Karnataka**
Member of Technical Staff - 2 (Promoted) — ***Jul 2021 - Jul 2024***

- Worked on a caption generation system prototype, leveraging **CLIP** & **GPT-3** alongside a supervised classification model.
- Spearheaded a significant feature enabling **immediate synchronization** of user edits, improving real-time collaboration.
- Led backend resiliency efforts achieving a **99.9%** service SLA across **three major Adobe Express releases**.
- **Optimized Node.js service memory usage by 60%** via low-level string and array buffer optimizations.

Technical Skills

Certifications: Introduction to Self-Driving Cars, Deep Learning, Advanced Learning Algorithms
Languages: Python, C, C++, JavaScript, TypeScript, SQL
Technologies & Frameworks: PyTorch, HuggingFace, Unsloth, vLLM, DeepSpeed, NumPy, Pandas, Tensorflow, PySpark
Tools & Platforms: ClearML, AWS, GCP (Dataproc), Git, Linux